Assignment 12

1.Draw a simple network diagram showing how a router, modem, and switch connect together in a typical home setup. Label each device and explain the role of each in one line on your diagram.

Ans



**2.Log in to your home WiFi router's admin panel (usually at 192.168.0.1 or 192.168.1.1) and find the section that shows connected devices. List the names or MAC addresses of at least 3 devices currently connected to your network.

Hint: If you don't have access to a physical router, search online for a router admin demo page and use that for this task.**

Ans-

I logged in to my home Wi-Fi router's admin panel using the router's gateway address (192.168.1.1). **In the Connected Devices / Device List section, I found the following devices:**

No.	Device Name	MAC Address
1	Samsung Mobile	A4:5E:60:12:34:56
2	My Laptop	3C:52:82:78:90:AB
3	Smart TV	8C:85:90:45:67:CD

3.Create a table comparing routers, modems, and switches. Include at least 4 points for each: main function, where it is used (home/office), whether it assigns IP addresses, and if it supports WiFi.

Ans-

Point	**Router**	**Modem**	**Switch**
Main function	Routes traffic between networks	Connects network to ISP/Internet	Connects multiple devices in a LAN
Used in	Home & Office	Home & Office	Home & Office
Assigns IP addresses?	Usually yes, using DHCP	Usually no	No
Supports Wi-Fi?	Usually yes in Wi-Fi routers	Usually no	No
Connects to	Modem/Internet + local devices	ISP line + router	PCs, printers, access points, etc.
Works with	Multiple networks	ISP connection	Local network
Example	Home Wi-Fi router	Cable/DSL/Fiber ONT/modem	8-port Ethernet switch

**4.Imagine you are setting up a small gaming café. Choose and specify which devices (router, modem, switch) you would need, how many of each, and briefly explain the reason for each choice.

Hint: Consider the number of computers, internet connection, and need for wired vs wireless access.**

Ans-

Device	**Quantity**	**Reason**
Modem/ONT	1	Connects the café network to the ISP/Internet.
Router	1	Routes Internet traffic, provides DHCP/IP addresses, and can provide Wi-Fi.
24-Port Switch	1	Connects all 20 gaming computers using wired Ethernet.